Mathematical Logic

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Spring 2026

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1 Propositional logic

1.1 Propositional formulas

Definition 1.1.1. A proposition is a statement which is either **true** (T) or **false** (F). Given propositional variables p and q we can combine these into longer propositions using the following connective, which are represented by truth tables:

p	q	$(\neg p)$	$(p \wedge q)$	$(p \vee q)$	$(p \rightarrow q)$	$(p \leftrightarrow q)$	$(p \downarrow q)$
T	T	F	T	T	T	T	F
T	F	F	F	T	F	F	F
F	T	T	F	T	T	F	F
F	F	T	F	F	F	T	T

All **propositional formulas** are either:

- propositional variables like p or q;

- the negation $(\neg\phi)$ for some propositional formula ϕ;
- a connective applied to two propositional formulas $(\phi \cdot \psi)$ for $\cdot \in \{\wedge, \vee, \rightarrow, \leftrightarrow, \downarrow\}$.

Because of the strict bracketing, we can use induction to prove every formula is either a propositional variable or build from strictly shorter formulas in a unique way.

Lemma 1.1.2. The connectives $\vee$ and $\wedge$ are associative, so we often omit brackets in expressions like $(p_1 \vee (p_2 \vee p_3))$ for $(p_1 \vee p_2 \vee p_3)$.

Proof. Truth tables, and induction on number of connectives. □

Definition 1.1.3. For $n \in \mathbb{N}$, a **truth function** of n variables is a function $f : \{T, F\}^n \rightarrow \{T, F\}$. Given a propositional formula ϕ with variables among $\{p_1, \ldots, p_n\}$, there is a associated truth function F_ϕ whose value at $(x_1, \ldots, x_n)$ is the truth value of ϕ when each p_i has value x_i.

Lemma 1.1.4. There are 2^{2^n} truth functions of n variables.

Proof. These are just functions, so we can count. □

Definition 1.1.5. A set of connectives S is called **adequate** if for all $n \geq 1$, for all truth functions f of n variables, there exists some formula which only involves connectives from S, and variables $p_1, \ldots, p_n$.

Theorem 1.1.6. The set $\{\neg, \wedge, \vee\}$ is adequate.

Proof. Let $G : \{T, F\}^n \rightarrow \{T, F\}$ be the truth function. If $G \equiv F$ on all inputs, then take $\phi = (p_1 \wedge (\neg p_1))$, and we have $G = F_\phi$.

Now write $\{v_1, \ldots, v_k\}$ for the set of all $v \in \{T, F\}^n$ such that $G(v) = T$. For each $v_i = (v_{i1}, \ldots, v_{il})$ we define:

$$q_{ij} = \begin{cases} p_j & \text{if } v_{ij} = T \\ (\neg p_j) & \text{if } v_{ij} = F \end{cases}$$

and then define $\psi_i := (q_{i1} \wedge \cdots \wedge q_{il})$, a propositional formula which is true iff each p_j has the same value as v_{ij}. Now take $\phi = (\psi_1 \vee \cdots \vee \psi_k)$, so we have $G = F_\phi$, where ϕ only consists of connectives in $\{\neg, \wedge, \vee\}$. □

The form of propositions used in this proof is **disjoint normal form** written dnf.

Corollary 1.1.7. Suppose χ is a formula whose truth function is not always F, then χ is logcially equivalent to a formula in disjunctive normal form.

Corollary 1.1.8. To show any other set of connectives S is adequate, we just have to construct the truth tables for $\{\neg, \wedge, \vee\}$ from connectives in S.

Corollary 1.1.9. The sets of connectives $\{\neg, \vee\}$, $\{\neg, \wedge\}$, $\{\neg, \rightarrow\}$, and $\{\downarrow\}$ are all adequate.

Definition 1.1.10. A propositional formula ϕ is called a **tautology** if its truth function always has value T. If ψ is another propositional formula, then we say ϕ and ψ are **logically equivalent** iff $(\phi \leftrightarrow \psi)$ is a tautology.

1.2 The formal system K

Definition 1.2.1. The **formal system** L (for propositional logic) has the following ingredients:

- An **alphabet** of symbols which consists of **variables** $\{p_1, p_2, \ldots\}$, **connectives** $\{\neg, \rightarrow\}$, and **punctuation** $\{(,)\}$.
- **Formulas**, which are certain finite sequences of symbols from the alphabet constructed as follows
  - any variable is a formula;
  - If ϕ and ψ are formulas of L, then so are $(\neg\phi)$ and $(\phi \rightarrow \psi)$
  - All formulas arise in this way.

  We call these L**-formulas**.

- **Axioms**, suppose ϕ, ψ, χ are L-formulas, then the following are **Axioms of** L:

  (A1) $(\phi \to (\psi \to \phi))$

  (A2) $((\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi)))$

  (A3) $(((\neg\psi) \to (\neg\phi)) \to (\phi \to \psi))$

- **Deduction rules**, in L this is just modus ponens

  (MP) From ϕ, and $(\phi \to \psi)$, deduce ψ.

The collection of an alphabet, formulas, axioms, and deduction rules, constitute a general **formal deduction system**.

Definition 1.2.2. A **proof** in a formal deduction system Σ is a finite sequence of Σ-formulas, each of which is either an axioms, or deduced from previous formulas in the proof by a deduction rule. The last formula in a proof is called a **theorem** of Σ. The number of formulas in the proof is called its **length**.

The main goal of this section is to show that the theorems of L are precisely the tautologies.

Definition 1.2.3. Suppose Γ is a set of L-formulas, then a **deduction from** Γ is a finite sequence of L-formulas, each of which is either an axioms, in Γ, or a deduction from the previous terms. Write $\Gamma \vdash_L \phi$ if there is a deduction from Γ ending in ϕ, in which case we say ϕ is a **consequence** of Γ.

The theorems of L are precisely the formulas ϕ such that $\emptyset \vdash_L \phi$, we normally write this as $\vdash_L \phi$.

Proposition 1.2.4 (Deduction theorem)**.** Suppose Γ is a set of L-formulas and ϕ, ψ are L-formulas such that $\Gamma \cup \{\phi\} \vdash_L \psi$, then $\Gamma \vdash_L (\phi \to \psi)$.

Proof. We go by induction on n, the length of the deduction $\Gamma \cup \{\phi\} \vdash_L \psi$.

If $n = 1$, then ψ is either an axiom, in Γ, or is ϕ. In the first two cases $\Gamma \vdash_L \psi$ is a one line deduction. Otherwise, showing $(\phi \to \phi)$ is a theorem is a classic exercise.

For the inductive step, in the deduction $\Gamma \cup \{\phi\} \vdash_L \psi$, then final line is either ψ, in Γ, or is ϕ, in which case we argue as in the inductive step. Otherwise ψ is obtained from earlier steps using MP, in which case there are formulas χ and $(\chi \to \psi)$ earlier on in the deduce, we apply the induced to step to get $\Gamma \vdash_L (\phi \to \chi)$ and $\Gamma \vdash_L (\phi \to (\chi \to \psi))$. Now (A2) and two uses of MP, gives $\Gamma \vdash_L (\phi \to \psi)$ as required. ☐

Corollary 1.2.5 (Hypothetic syllogism)**.** Suppose ϕ, ψ, χ are L-formulas, such that $\vdash_L (\phi \to \psi)$ and $\vdash_L (\psi \to \chi)$, then $\vdash_L (\phi \to \chi)$.

Proof. We use the deduction theorem for $\Gamma = \emptyset$ on $\vdash_L (\phi \to \psi)$, and modus ponenes on the two theorems we have to deduce χ, now use the converse of the deduction theorem to deduce $\phi \to \chi$. ☐

I will frequently write just $\vdash$ for $\vdash_L$.

1.3 Soundness

Definition 1.3.1. A **propositional valuation** is an assignment of truth values to the propositional variable $\{p_1, p_2, \ldots\}$ which we can view as a function $v : \mathbb{N} \to \{T, F\}$. Using truth tables, we can extend v to a function on the set of L-formulas. This will have some immediate properties such as $v((\neg\phi)) \neq v(\phi)$ and $v((\phi \to \psi)) = F$ iff $v(\phi) = T$ and $v(\psi) = F$.

Valutations correspond to rows in a truth table.

Theorem 1.3.2 (Soundness of L)**.** Suppose ϕ is a theorem of L, then ϕ is a tautology.

Proof. We do induction on the length of a proof of $\vdash \phi$.

In the base case, ϕ must be an axiom, by truth tables three axioms are tautologies.

In the inductive step, we assume ϕ has a proof of length n, the last step is either: an axiom, in which case we argue as in the base case; or modus ponens from two theorems χ and $(\chi \to \phi)$, by induction we know both of these are tautologies, and now by considering the truth tables, we know χ and $(\chi \to \phi)$ are both true implies ϕ is true. ☐

Using the same proof strategy, there is an immediate generalisation.

Corollary 1.3.3 (Generalisation of soundness)**.** Suppose Γ is a set of L formulas and ϕ is an L-formula such that $\Gamma \vdash \phi$. Suppose v is a valuation with $v(\Gamma) = \{T\}$, then $v(\phi) = T$.

1.4 Completeness

Definition 1.4.1. Suppose Γ is a set of L-formulas. We say Γ is **consistent** if there is no L-formula ϕ such that $\Gamma \vdash \phi$ and $\Gamma \vdash (\neg\phi)$. We say Γ is **complete** is for every L-formula ϕ either $\Gamma \vdash \phi$ or $\Gamma \vdash (\neg\phi)$.

Unfortunately, the completeness theorem says in the case $\Gamma = \emptyset$, Γ is consistent. But in this case, Γ is not complete. The word complete is being used with two *completely* different meanings.

Proposition 1.4.2. Suppose Γ is a consistent set of L-formulas and $\Gamma \nvdash \phi$, then $\Gamma \cup \{(\neg\phi)\}$ is consistent.

Proof. Assum for a contradiction there exists some L-formula ψ such that

$$\Gamma \cup \{(\neg\phi)\} \vdash \psi,\ (\neg\psi)$$

by passing the second statement through the deduction formula we get $\Gamma \vdash ((\neg\phi) \to (\neg\psi))$, we then use (A3) to get $\Gamma \vdash (\psi \to \phi)$, we can now use the first statement to get $\Gamma \cup \{(\neg\phi)\} \vdash \phi$. Note that for any L-formula χ, the formula $(((\neg\chi) \to \chi) \to \chi)$ is a theorem of L, so we get $\Gamma \vdash \phi$, which contradicts the assumption, thus $\Gamma \cup \{(\neg\psi)\}$ is consistent. □

Proposition 1.4.3 (Lindenbaum lemma)**.** Suppose Γ is a consistent set of L-formulas. Then there is a set of L-formulas $\Gamma^* \supset \Gamma$ which is consistent and complete.

Proof. The set of L-formulas is countable, so we can list them as $\phi_0, \phi_1, \phi_2, \dots$ and inductively define a chain of sets of formulas:

$$\Gamma_0 \subseteq \Gamma_1 \subseteq \cdots, \qquad \Gamma_0 = \Gamma, \qquad \Gamma_{n+1} = \begin{cases} \Gamma_n & \text{if } \Gamma_n \vdash \phi_n \\ \Gamma_n \cup \{(\neg\phi_n)\} & \text{if } \Gamma_n \nvdash \phi_n \end{cases}$$

Now by induction with the previous proposition, each of the Γ_i is consistent. We let

$$\Gamma^* = \bigcup_{i \in \mathbb{N}} \Gamma^i$$

and claim Γ^* is consistent. If $\Gamma^* \vdash \phi, (\neg\phi)$, then as deduction have finite length, there exists some n with $\Gamma_n \vdash \phi, (\neg\phi)$, a contradiction. To show Γ^* is complete, suppose χ is such that $\Gamma^* \nvdash \chi$, there is some n with $\phi_n = \chi$ and so $\Gamma_n \nvdash \chi$, thus $(\neg\chi) \in \Gamma_{n+1}$, and therefore $\Gamma^* \vdash (\neg\chi)$. □

Lemma 1.4.4. Suppose Γ^* is a complete, consistent set of L-formulas. Then there is a valuation v such that, for every L-formula ϕ, $v(\phi) = T$ iff $\Gamma^* \vdash \phi$.

Proof. As Γ^* is complete, we know for every propositional variable p_i, either $\Gamma^* \vdash p_i$ or $\Gamma^* \vdash (\neg p_i)$. Let v be the unique valuation such that $v(p_i) = T$ iff $\Gamma^* \vdash p_i$. We will no do induction on the length of a formula ϕ, to show v satisfies our desired property, in the base case this is by definition. For the inductive step, there are two possible cases.

Case 1: ϕ is $(\neg\psi)$ for some ψ. Here, $v(\phi) = T$ iff $v(\psi) = F$ iff $\Gamma^* \nvdash \psi$ iff $\Gamma^* \vdash (\neg\psi)$ iff $\Gamma^* \vdash \phi$.

Case 2: ϕ is $(\psi \to \chi)$. Here, $v(\phi) = F$ iff $v(\psi) = T$ and $v(\chi) = F$ iff $\Gamma^* \vdash \psi$ and $\Gamma^* \nvdash \chi$. If $\Gamma^* \vdash \phi$ then we could do modus ponens to get $\Gamma^* \vdash \chi$ which contradicts consistency. If $\Gamma^* \nvdash \phi$ then $\Gamma^* \nvdash \chi$, otherwise by (A1) we get $\vdash (\chi \to (\psi \to \chi))$; similarly if $\Gamma^* \vdash (\neg\psi)$, then by the problem sheet we know $\vdash ((\neg\psi) \to (\psi \to \chi))$, thus by the inductive hypothesis $v(\chi) = F$ and $v((\neg\psi)) = F$, thus $v(\phi) = F$. □

Corollary 1.4.5. Suppose Γ is a consistent set of L-formulas, then there is a valuation v with $v(\Gamma) = T$.

Corollary 1.4.6. Suppose Δ is a set of L-formulas which is consistent and $\Delta \nvdash \phi$, then there is a valuation with $v(\Delta) = T$ and $v(\phi) = F$.

Theorem 1.4.7 (Generalised completeness of L)**.** If Δ is a consistent set of L-formulas, and ϕ is an L-formula such that for every valuation v with $v(\Delta) = T$, we have $v(\phi) = T$, then $\Delta \vdash \phi$.

Proof. This is the contrapositive of the previosu corrolary. □

Corollary 1.4.8 (Completeness of L)**.** If the L-formula ϕ is a tautology, then $\vdash \phi$.

Theorem 1.4.9 (Compactness theorem for L)**.** Suppose Γ is a set of L-formulas, then tfae:

1. There is a valuation v with $v(\Gamma) = T$

2. For every finite subset $\Sigma \subseteq \Gamma$, there is a valuation w with $w(\Sigma) = T$.

Proof sketch. (1) definitely implies (2), so suppose (1) doesn't hold. By the first corrolary, Γ isn't consistent. As deductions are finite, there is a finite subset $\Sigma \subseteq \Gamma$ which is also inconsistent, then by the generalised soundness theorem, the desired valuation of (2) cannot exist. $\square$

2 First-order logic

2.1 Structures

Definition 2.1.1. Suppose A is a set and $n \geq 1$: an n**-ary relation** on A is a subset $\overline{R} \subseteq A^n$; an n**-ary function** on A is a function $\overline{f} : A^n \to A$.

Some examples are: equality and orders as 2-ary relationships, various binary operations as 2-ary functions, subsets as 1-ary relations. In future for $\overline{R} \subseteq A^n$ and $(a_1, \ldots, a_n) \in A^n$ we write $\overline{R}(a_1, \ldots, a_n)$ for $(a_1, \ldots, a_n) \in \overline{R}$.

Definition 2.1.2. A **first-order structure** $\mathcal{A}$ consists of:

- a nonemtpy set A, called the **domain** of $\mathcal{A}$;
- a set $\{\overline{R}_i \mid i \in I\}$ of **relations** on A, $\overline{R}_i \subseteq A^{n_i}$ for $i \in I$;
- a set $\{\overline{f}_j \mid j \in J\}$ of **functions** $\overline{f}_j : A^{m_j} \to A$ for $j \in J$;
- a set $\{\overline{c}_k \mid k \in K\} \subseteq A$ of **constants**.

The **indexing sets** I, J, K can be empty, and are normally subsets of $\mathbb{N}$. The information:

$$(n_i \mid i \in I) \qquad (m_j \mid j \in J) \qquad K$$

is called the **signature** of $\mathcal{A}$. We will often write

$$\mathcal{A} = \big\langle A; (\overline{R}_i \mid i \in I), (\overline{f}_j \mid j \in J), (\overline{c}_k \mid k \in K) \big\rangle$$

Now for some more fleshed out examples:

- Any ordered set A (e.g. $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$) can be a first order structure with $I = \{1\}$, $J = K = \emptyset$ and $\overline{R}_1(a_1, a_2)$ to mean $a_1 < a_2$.
- Any group G with a binary relation $\overline{R}$ for equality, a binary function $\overline{m}$ for multiplication, a unary function $\overline{i}$ for inversion, and the constant element $\overline{e}$.
- Any ring R with a binary relation $\overline{R}$ for equality, two binary functions $\overline{m}$ and $\overline{e}$ for multiplication and addition respectively, the unary function $\overline{n}$ for negation, and the constants $\overline{0}$ and $\overline{1}$.
- Peano arithmetic $\big\langle \mathbb{N}; \overline{R}, \overline{m}, \overline{a}, \overline{s}, \overline{0} \big\rangle$ where $\overline{R}, \overline{m}, \overline{a}, \overline{0}$ are as in a ring, and $\overline{s}$ is the successor function $\overline{s}(n) = n + 1$.
- Graphs with two binary relations $\overline{R}$ for equality and $\overline{E}$ for adjacency.

Definition 2.1.3. A **first-order language** $\mathcal{L}$ has an alphabet of symbols:

variables	$x_0, x_1, \ldots$
connectives	$\neg, \to$
punctuation	$($, $)$
quantifier	$\forall$
relation symbols	R_i for $i \in I$
function symbols	f_j for $j \in J$
constant symbols	c_k for $k \in K$

where I, J, K are indexing sets, which could be empty. Each symbol R_i or f_j comes equipped with an arity n_i or m_j respectively, and the signature of $\mathcal{L}$ can be defined as with structures.

An $\mathcal{L}$**-structure** is a structure $\mathcal{A}$ with the same signature as $\mathcal{L}$, there is a correspondence between relation, function, and constant symbols of $\mathcal{L}$ with the actual relations, functions, and constants of $\mathcal{A}$ such that arities agree. This correspondence is called an **interpretation** of $\mathcal{L}$.

In our previous examples, we have given the first-order languages of groups, rings, etc where the actual mathematical objects are exactly the $\mathcal{L}$-structure for these languages.

Definition 2.1.4. A **term** of $\mathcal{L}$ is defined as follows:

- any variable is a term;
- any constant symbol is a term;
- if f is an m-ary function symbol of $\mathcal{L}$ and $t_1, \ldots, t_m$ are terms, then $f(t_1, \ldots, t_m)$ is also a term;
- all terms arise in this way.

An **atomic formula** of $\mathcal{L}$ is of the form $R(t_1, \ldots, t_n)$ where R is an n-ary relation symbol of $\mathcal{L}$ and $t_1, \ldots, t_n$ are terms.

An **$\mathcal{L}$-formula** is defined as follows:

- an atomic formula is an $\mathcal{L}$-formula;
- If ϕ and ψ are $\mathcal{L}$-formulas, then $(\neg\phi)$, $(\phi \to \psi)$, and $(\forall x)\phi$ are all $\mathcal{L}$-formulas, where x is any variable;
- every $\mathcal{L}$-formula arises in this way.

Definition 2.1.5. With the same notation as above, if $\mathcal{A}$ is an $\mathcal{L}$-structure, a **valuation** in $\mathcal{A}$ is a function v from the set of terms of $\mathcal{L}$ to A such that for all $k \in K$, $v(c_k) = \overline{c}_k$; and for all terms $t_1, \ldots, t_m$ and m-ary function symbols f, $v(f(t_1, \ldots, t_m)) = \overline{f}(v(t_1), \ldots, v(t_m))$.

Lemma 2.1.6. Suppose $\mathcal{A}$ is an $\mathcal{L}$-structure and $a_0, a_1, \ldots \in A$, then there is a unique valuation v in $\mathcal{A}$ with $v(x_l) = a_l$ for all $l \in \mathbb{N}$.

Proof. By induction on the length of the terms, if we make v satisfy everything, then we can build v for all of the terms. □

Definition 2.1.7. Suppose $\mathcal{A}$ is an $\mathcal{L}$-structure, and x_l is a variable. If v, w are valuations in $\mathcal{A}$ such that for all $m \neq l$, $v(x_m) = w(x_m)$, then we say v and w are **x_l-equivalent**.

Definition 2.1.8. Suppose $\mathcal{A}$ is an $\mathcal{L}$-structure, and v is a valuation in $\mathcal{A}$, then for all $\mathcal{L}$-formulas ϕ, define recursively v **satisfues** ϕ, written $v[\phi] = T$, in $\mathcal{A}$ if:

- if R is an n-ary relation symbol and $t_1, \ldots, t_n$ are terms of $\mathcal{L}$, then v satisfies $R(t_1, \ldots, t_n)$ iff $\overline{R}(v(t_1), \ldots, v(t_n))$ holds in $\mathcal{A}$;
- if ϕ and ψ are $\mathcal{L}$-formulas and we know if v satisfies these, then:
  - v satisfies $(\neg\phi)$ in $\mathcal{A}$ iff v does not satisfy ϕ in $\mathcal{A}$,
  - v does not satisfy $(\phi \to \psi)$ in $\mathcal{A}$ iff v satisfies ϕ in $\mathcal{A}$ but v doesn't satisfy ψ in $\mathcal{A}$,
  - v satisfies $(\forall x_l)\phi$ in $\mathcal{A}$ iff for all valuations w in $\mathcal{A}$ such that w is x_l-equivalent to v, then w satisfies ϕ in $\mathcal{A}$.

If every valuation in $\mathcal{A}$ satisfies the $\mathcal{L}$-formula ϕ we say ϕ is **true** in $\mathcal{A}$ or $\mathcal{A}$ is a **model** of ϕ and write $\mathcal{A} \models \phi$. If $\mathcal{A} \models \phi$ for every $\mathcal{L}$-structure $\mathcal{A}$, we say ϕ is **logically valid** and write $\models \phi$.

Logically valid formulas are clearly the tautologies of first-order languages. While there is an algorithm, by truth tables, to determine if any propositional formula is a tautology, Gödel's incompleteness theorem tells us there is no such algorithm for a general $\mathcal{L}$-formula.

Definition 2.1.9. Suppose ϕ and ψ are $\mathcal{L}$-formulas, then we will use the notation $(\exists x)\phi$ for $(\neg(\forall x)(\neg\phi))$ and $(\phi \vee \psi)$ for $((\neg\phi) \to \psi)$ and the other propositional logic symbols.

Lemma 2.1.10. Suppose $\mathcal{A}$ is an $\mathcal{L}$-structure and ϕ is an $\mathcal{L}$-formula. Let v be a valuation in $\mathcal{A}$, then v satisfues $(\exists x_1)\phi$ in $\mathcal{A}$ iff there is a valuation w, which is x_1-equivalent ot v, such that w satisfies ϕ.

Proof. Suppose v satisfies $(\neg(\forall x_1)(\neg\phi))$, then v does not satisfy $(\forall x_1)(\neg\phi)$. So there is a valuation w, x_1-euquivalent to v, such that w does not satisfy $(\neg\phi)$. Such a valuation w satisfies ϕ. □

Definition 2.1.11. Suppose χ is an $\mathcal{L}$-formula involving the propositional variables $p_1, \dots, p_n$. Suppose $\mathcal{L}$ is a first-order language and $\phi_1, \dots, \phi_n$ are $\mathcal{L}$-formulas. A **substitution instance** of χ is obtained by replacing each p_i in χ by ϕ_i, call the resulting $\mathcal{L}$-formula θ.

Theorem 2.1.12. With the above notation, we have the following:

1. θ is an $\mathcal{L}$-formula;
2. Suppose v is a valuation in an $\mathcal{L}$-structure $\mathcal{A}$, and w is the propositional valuation with $w(p_i) = v[\phi_i]$ for all i, then $v[\theta] = w(\chi)$;
3. if χ is a tautology, then θ is logically valid.

Proof. We omit the proof of (1), as (3) follows from (2), proving (2) is all that remains, and this follows exactly from induction on the length of χ. $\square$

There are many logically valid formulas which don't arise in this way.

2.2 Bound and free variables

Definition 2.2.1. suppose ϕ and ψ are $\mathcal{L}$-formulas and $(\forall x_i)\phi$ occurs as a subformula of ψ, then we say

- ϕ is in the **scope** of the quantifier $(\forall x_i)$. An occurence of a variable x_j in ψ is **bound** if it is in the scope of a quantifier $(\forall x_j)$ in ψ.
- Otherwise, it is a **free** occurence of x_j, variables have a free occurrence in ψ are called the **free variables** of ψ.
- A formula with no free variables is called a **closed formula** or a **sentence** of $\mathcal{L}$.

Definition 2.2.2. If ψ is an $\mathcal{L}$-formula with free variabels amongst $x_1, \dots, x_n$, then we may write $\psi(x_1, \dots, x_n)$ for ϕ. If $t_1, \dots, t_n$ are terms, then by $\psi(t_1, \dots, t_n)$ we mean the $\mathcal{L}$-formula obtained by replacing each **free** occurence of x_i in ψ by t_i.

Proposition 2.2.3. Suppose ϕ is a closed $\mathcal{L}$-formula and $\mathcal{A}$ is an $\mathcal{L}$-structure, then either $A \vDash \phi$ or $\mathcal{A} \vDash (\neg\phi)$. More generally, if ϕ has free variables amongs $x_1, \dots, x_n$ and v, w are valuations in $\mathcal{A}$ with $v(x_i) = w(x_i)$ for all i, then $v[\phi] = T \iff w[\phi] = T$.

Proof. Note that the first statement follows from the generalisation with $n = 0$. We will do induction on the number of connectives and quantifiers in ϕ.

For the base case ϕ is atomic, so we let $\phi = R(t_1, \dots, t_m)$ where there t_j are terms. The t_j only involve variables amongst $x_1, \dots, x_n$, now as v and w agree on all these, we will have

$$v[R(t_1, \dots, x_n)] = T \iff \overline{R}(v(t_1), \dots, v(t_m)) \iff \overline{R}(w(t_1), \dots, w(t_m)) \iff w[R(t_1, \dots, x_n)] = T$$

For the inductive step, ϕ is either $(\neg\psi)$, $(\psi \to \chi)$ or $(\forall x_i)\psi$, the first two cases are left as exercises, we will do the third. If $v[\phi] = F$, then were is a valuation v' which is x_i-equivalent to v with $v'[\psi] = F$. Let w' be the valuation x_i-equivalent to w with $w'(x_i) = v'(x_i)$, then v' and w' agree on the free variables $x_1, \dots, x_n$ of ψ, so by the inductive hypothesis we get $v'[\psi] = w'[\psi]$. So if $w'[\psi] = F$, and thus $w[\phi] = F$. $\square$

If $\mathcal{A}$ is an $\mathcal{L}$-structure and $\psi(x_1, \dots, x_n)$ is an $\mathcal{L}$-formula, with $a_1, \dots, a_n \in A$, the domain of $\mathcal{A}$, then we write $\mathcal{A} \vDash \psi(a_1, \dots, a_n)$ to mean $v[\psi] = T$ for every valutation v in $\mathcal{A}$ with $v(x_i) = a_i$ for all i.

2.3 Substituting terms for variables

Definition 2.3.1. Let ϕ be an $\mathcal{L}$-formula, x_i a variable, and t an $\mathcal{L}$-term, we say t is **free for** x_i **in** ϕ if there is no variable x_j in t such that x_i has a free occurrence within the scope of a quantifier $(\forall x_j)$ in ϕ.

Theorem 2.3.2. Suppose $\phi(x_1)$ is an $\mathcal{L}$-formula, possible with other free variables, and let t be a term free for x_1 in ϕ, then $\vDash ((\forall x_1)\phi(x_1) \to \phi(t))$.

Proof. As per usual, we will go by induction on the number of connectives and quantifiers in ϕ.

For the base case, ϕ is an atomic formula $R(u_1, \dots, u_m)$. Let u_i^* be the result of substituting t for x_1 in u_i. Then by induction on the length of the terms, each u_i^* is a term, and $v'(u_i) = v(u_i^*)$ $\square$